Experiment-9

Develop an Android Calculator Application using Android Studio

AIM:

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

ALGORITHM:

1. **Initialize the Android Application:**
   - Set up a new Android project in **Android Studio**.
   - Create a **MainActivity** Java or Kotlin file for the main screen of the application.
2. **Design the User Interface (UI):**
   - Create the following UI components in the **XML layout** file:
     - **EditText** for input fields where the user will enter numbers (two EditTexts for number1 and number2).
     - **Buttons** for each arithmetic operation (Addition, Subtraction, Multiplication, Division).
     - **TextView** to display the result of the operation.
3. **Declare Variables:**
   - Create variables for the **EditText** fields, **Buttons**, and **TextView**.
   - Define variables to hold the input numbers and the result (e.g., num1, num2,result)
4. **Get Input from the User:**
   - Get the values entered by the user in **EditText** fields using getText().toString() and convert them to numerical values (like Integer.parseInt() or Double.parseDouble()).
5. **Set Event Listeners for Buttons:**
   - For each button (Addition, Subtraction, Multiplication, and Division), set an **OnClickListener**.
   - Inside the **onClick()** method for each button, perform the respective arithmetic operation:
     - **Addition**: result = num1 + num2

- **Subtraction**: result = num1 - num2
- **Multiplication**: result = num1 * num2
- **Division**: result = num1 / num2 (add validation to prevent division by zero).

6. **Perform Arithmetic Operation:**
   - Check which button was clicked (using view.getId()), and based on the button, perform the respective arithmetic operation.
   - **Handle Division by Zero**: Check if num2 is 0 before performing division and show an appropriate error message if true.
7. **Display the Result:**
   - Use **TextView** to display the result of the operation.
   - Convert the result to a string and set it using setText() method on the **TextView**.
8. **Validate User Inputs:**
   - Before performing any operation, validate the input to ensure the user has entered valid numbers (not empty or non-numeric characters).
9. **Handle Edge Cases:**
   - Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.
10. **Finish the Operation:**
    - Once the result is displayed, the user can either perform another calculation or exit the app.

CODE:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Simple Calculator"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:layout_marginTop="50dp"
        android:textSize="30sp"
        />
    <EditText
        android:id="@+id/number1"
        android:hint="Enter a number 1"
        android:inputType="number"
        android:layout_marginTop="50dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        />
    <EditText
        android:id="@+id/number2"
        android:hint="Enter a number 2"
        android:inputType="number"
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        />

    <LinearLayout
        android:layout_marginTop="30dp"
        android:orientation="horizontal"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">
        <Button
            android:id="@+id/btn_add"
            android:text="+"
            android:textSize="30sp"
            android:layout_marginEnd="5dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            />
        <Button
            android:id="@+id/btn_sub"
            android:text="-"
            android:textSize="30sp"
            android:layout_marginEnd="5dp"
            android:layout_width="wrap_content"
            android:layout_height="match_parent"
            />
        <Button
            android:id="@+id/btn_mul"
            android:text="x"
            android:textSize="30sp"
            android:layout_marginEnd="5dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            />
        <Button
            android:id="@+id/btn_div"
```

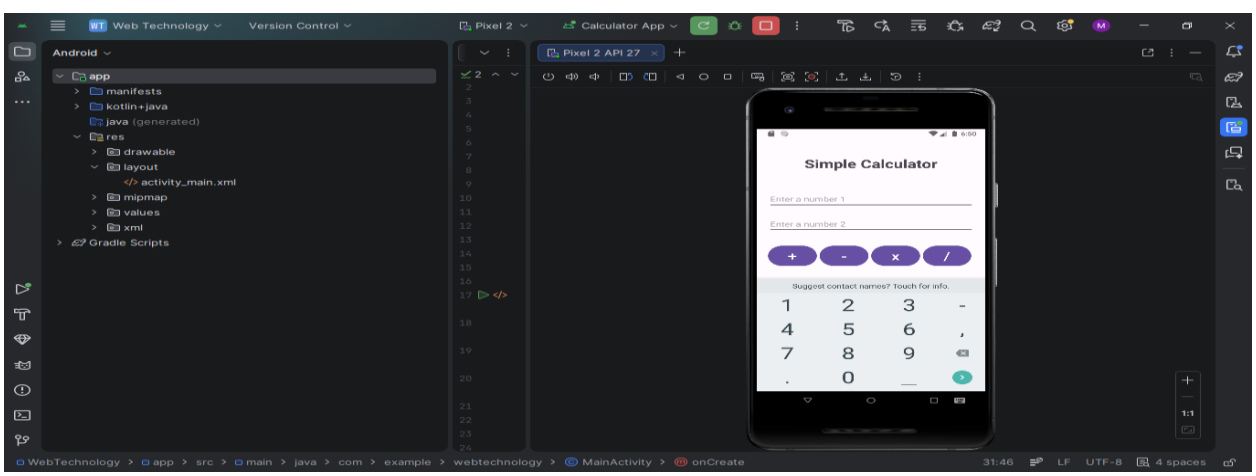
```
            android:text="/"
            android:textSize="30sp"
            android:layout_marginEnd="5dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            />

    </LinearLayout>
    <TextView
        android:id="@+id/answer"
        android:layout_gravity="center"
        android:layout_marginTop="30dp"
        android:textSize="30sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

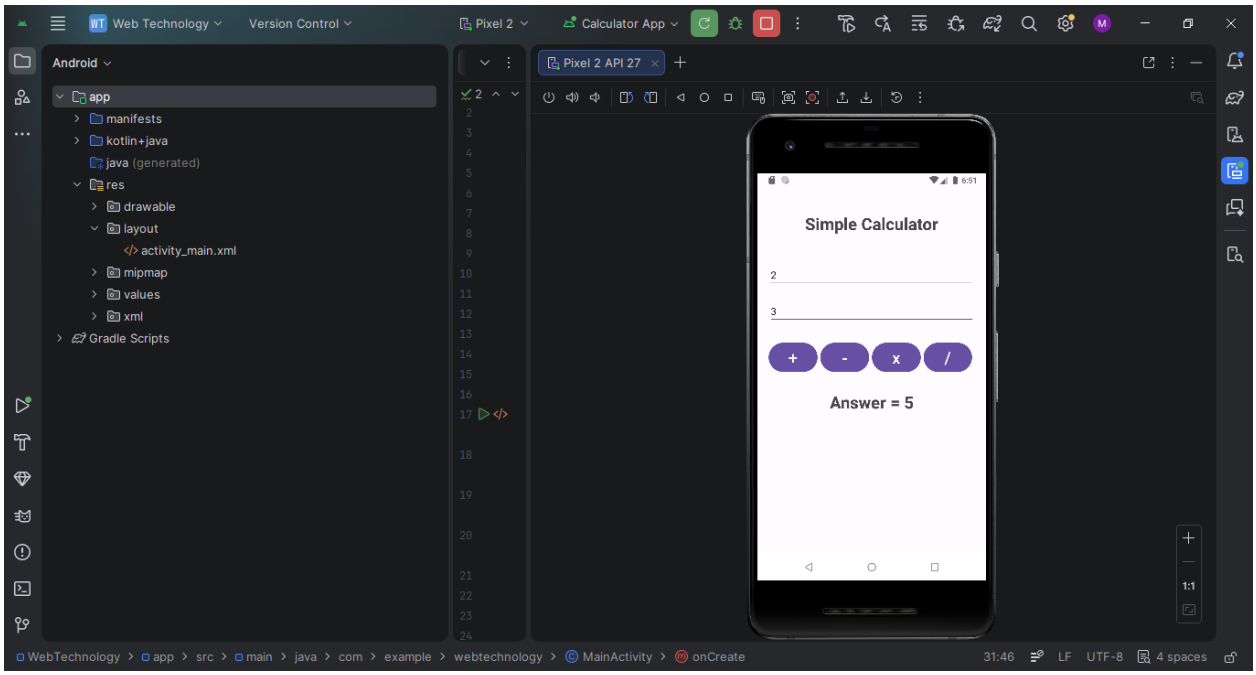
MainActivity.xml:

```
package com.example.webtechnology;

import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity implements View.OnClickListener{
    Button buttonAdd,buttonSub,buttonMul,buttonDiv;
    EditText editTextn1,editTextn2;
    TextView textview;
    int num1,num2;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        buttonAdd=findViewById(R.id.btn_add);
        buttonSub=findViewById(R.id.btn_sub);
        buttonMul=findViewById(R.id.btn_mul);
        buttonDiv=findViewById(R.id.btn_div);
        editTextn1=findViewById(R.id.number1);
        editTextn2=findViewById(R.id.number2);
        textview=findViewById(R.id.answer);
```

```java
        buttonAdd.setOnClickListener(this);
        buttonSub.setOnClickListener(this);
        buttonMul.setOnClickListener(this);
        buttonDiv.setOnClickListener(this);
    }
    public int getIntFromEditText(EditText editText){
        if(editText.getText().toString().equals(" ")){
            Toast.makeText(this,"Enter number",Toast.LENGTH_SHORT).show();;
            return 0;
        }
        else{
            return Integer.parseInt(editText.getText().toString());
        }


    }

    @Override
    public void onClick(View v) {
        num1=getIntFromEditText(editTextn1);
        num2=getIntFromEditText(editTextn2);
        int id = v.getId();

        if (id == R.id.btn_add) {
            textview.setText("Answer = " + (num1 + num2));
        }
        else if (id == R.id.btn_sub) {
            textview.setText("Answer = " + (num1 - num2));
        }
        else if (id == R.id.btn_mul) {
            textview.setText("Answer = " + (num1 * num2));
        }
        else if (id == R.id.btn_div) {
            textview.setText("Answer = " + ((float) num1 / num2));
        }
    }
}
```

Output:





RESULT:

An calculator is designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.